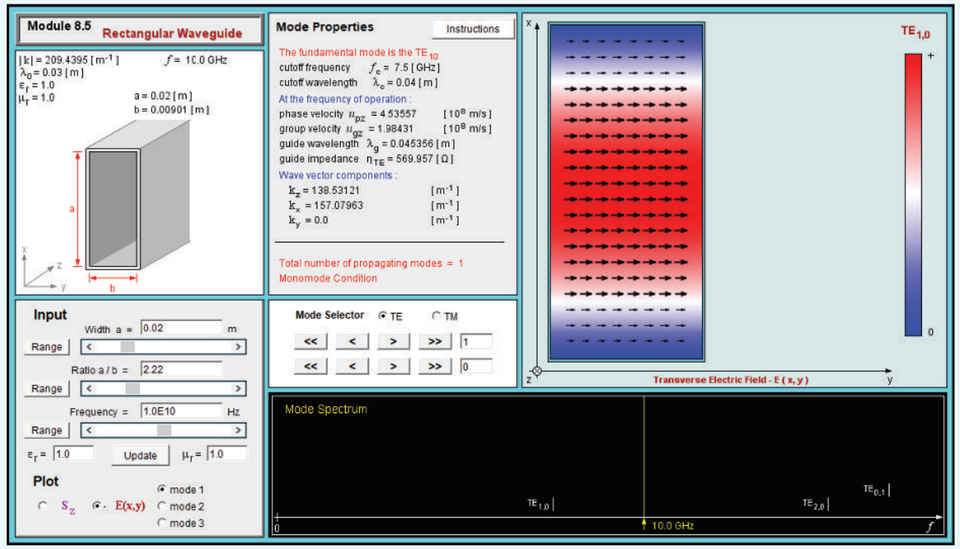
Module 8.5 Rectangular Waveguide Upon specifying the waveguide dimensions, the frequency f, and the mode type (TE or TM) and number, this module provides information about the wave impedance, cutoff frequency, and other wave attributes. It also displays the electric and magnetic field distributions inside the guide.



▶ The quality factor is defined in terms of the ratio of the energy stored in the cavity volume to the energy dissipated in the cavity walls through conduction. ◀

For an ideal cavity with perfectly conducting walls, no energy loss is incurred, as a result of which Q is infinite and $\Delta f \approx 0$. Metals have very high (but not infinite) conductivities, so a real cavity with metal walls stores most of the energy coupled into it in its volume. However, it also loses some energy to heat conduction. A typical value for Q is on the order of 10,000, which is much higher than can be realized with lumped RLC circuits.

Example 8-11: Q of a Resonant Cavity

The quality factor for a hollow resonant cavity operating in the TE_{101} mode is

$$Q = \frac{1}{\delta_s} \frac{abd(a^2+d^2)}{[a^3(d+2b)+d^3(a+2b)]}, \qquad (8.124)$$

where $\delta_s = 1/\sqrt{\pi f_{mnp}\mu_0\sigma_c}$ is the skin depth and σ_c is the conductivity of the conducting walls. Design a cubic cavity with a TE_{101} resonant frequency of 12.6 GHz and evaluate its bandwidth. The cavity walls are made of copper.

Solution: For $a = b = d$, $m = 1$, $n = 0$, $p = 1$, and $u_{p_0} = c = 3 \times 10^8$ m/s, Eq. (8.122) simplifies to

$$f_{101} = \frac{3\sqrt{2} \times 10^8}{2a} \quad \text{(Hz)}.$$

For $f_{101} = 12.6$ GHz, this gives

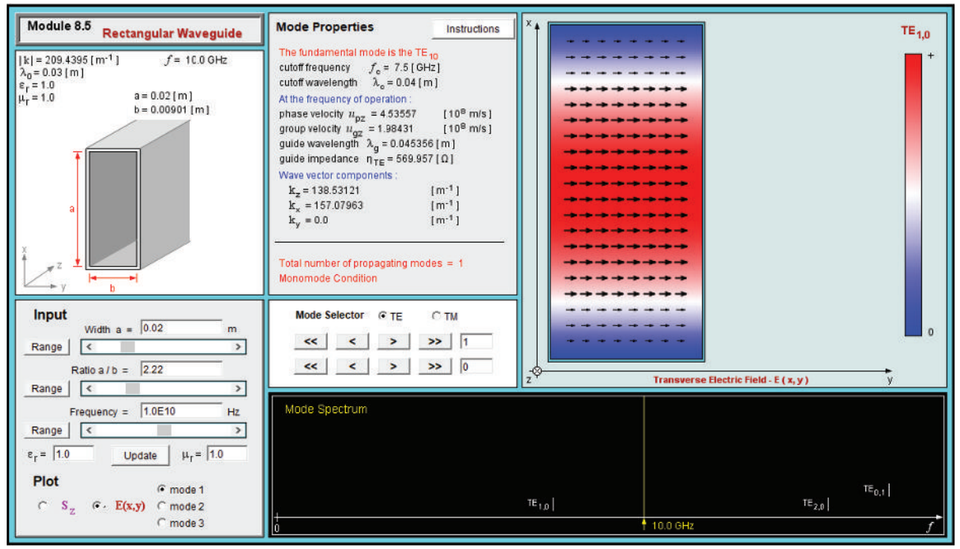
$$a = 1.68 \text{ cm}.$$

At $f_{101} = 12.6$ GHz, the skin depth for copper (with $\sigma_c = 5.8 \times 10^7$ S/m) is

$$\delta_s = \frac{1}{[\pi f_{101}\mu_0\sigma_c]^{1/2}} = \frac{1}{[\pi \times 12.6 \times 10^9 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7]^{1/2}} = 5.89 \times 10^{-7} \text{ m}.$$

376 CHAPTER 8 WAVE REFLECTION AND TRANSMISSION

模块 8.5 矩形波导 在指定波导尺寸、频率 f 以及模式类型（TE 或 TM）和数量后，此模块提供有关波阻抗、截止频率和其他波属性的信息。它还显示导轨内的电场和磁场分布。



▶ 品质因数是根据腔体积中存储的能量与通过传导在腔壁中耗散的能量的比率来定义的。 ◀

对于具有理想导电壁的理想空腔，不会产生能量损失，因此 Q 无穷大且 Δ f ≈ 0。金属具有非常高（但不是无限）的电导率，因此具有金属壁的真实空腔将耦合到其体积中的大部分能量存储起来。然而，它也会因焦耳热而损失一些能量。 Q 的典型值约为 10,000，该值远高于集总 RLC 电路所能实现的值。

例 8-11： 谐振腔的 Q 值

在 TE_{101} 模式下工作的空心谐振腔的品质因数为

$$Q = \frac{1}{\delta_s} \frac{abd(a^2+d^2)}{[a^3(d+2b)+d^3(a+2b)]}, \qquad (8.124)$$

其中 $\delta_s = 1/\sqrt{\pi f_{mnp}\mu_0\sigma_c}$ 是趋肤深度，σ_c 是导电壁的电导率。设计一个 TE_{101} 谐振频率为 12.6 GHz 的立方腔并评估其带宽。空腔壁由铜制成。

解：对于 a = b = d、m = 1、n = 0、p = 1 和 $u_{p_0} = c = 3 \times 10^8$ m/s，方程： (8.122) 简化为

$$f_{101} = \frac{3\sqrt{2} \times 10^8}{2a} \quad \text{(Hz)}.$$

对于 $f_{101} = 12.6$ GHz，这给出

$$a = 1.68 \text{ cm}.$$

在 $f_{101} = 12.6$ GHz 时，铜的趋肤深度（$\sigma_c = 5.8 \times 10^7$ S/m）为

$$\delta_s = \frac{1}{[\pi f_{101}\mu_0\sigma_c]^{1/2}} = \frac{1}{[\pi \times 12.6 \times 10^9 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7]^{1/2}} = 5.89 \times 10^{-7} \text{ m}.$$

Upon setting $a = b = d$ in Eq. (8.124), the expression for Q of a cubic cavity becomes

$$Q = \frac{a}{3\delta_s} = \frac{1.68 \times 10^{-2}}{3 \times 5.89 \times 10^{-7}} \approx 9{,}500.$$

Hence, the cavity bandwidth is

$$\Delta f \approx \frac{f_{101}}{Q} \approx \frac{12.6 \times 10^9}{9{,}500} \approx 1.3 \text{ MHz}.$$

Chapter 8 Summary

Concepts

- The relations describing the reflection and transmission behavior of a plane EM wave at the boundary between two different media are the consequence of satisfying the conditions of continuity of the tangential components of **E** and **H** across the boundary.
- Snell's laws state that $\theta_i = \theta_r$ and

$$\sin\theta_t = (n_1/n_2)\sin\theta_i.$$

  For media such that $n_2 < n_1$, the incident wave is reflected totally by the boundary when $\theta_i \geq \theta_c$, where θ_c is the critical angle given by $\theta_c = \sin^{-1}(n_2/n_1)$.
- By successive multiple reflections, light can be guided through optical fibers. The maximum data rate of digital pulses that can be transmitted along optical fibers is dictated by modal dispersion.
- At the Brewster angle for a given polarization, the incident wave is transmitted totally across the boundary. For nonmagnetic materials, the Brewster angle exists for parallel polarization only.
- Any plane wave incident on a plane boundary can be synthesized as the sum of a perpendicularly polarized wave and a parallel polarized wave.
- Transmission-line equivalent models can be used to characterize wave propagation, reflection by, and transmission through boundaries between different media.
- Waves can travel through a metal waveguide in the form of transverse electric (TE) and transverse magnetic (TM) modes. For each mode, the waveguide has a cutoff frequency below which a wave cannot propagate.
- A cavity resonator can support standing waves at specific resonant frequencies.

Important Terms

Provide definitions or explain the meaning of the following terms:

ω-β diagram
acceptance angle θ_a
angles of incidence, reflection, and transmission
Brewster angle θ_B
cladding
critical angle θ_c
cutoff frequency f_{mn}
cutoff wavenumber k_c
dominant mode
evanescent wave
fiber core
grazing incidence
group velocity u_g
index of refraction n
modal dispersion
modes
optical fibers
parallel polarization
perpendicular polarization
phase-matching condition
plane of incidence
polarizing angle
quality factor Q
reflection coefficient Γ
reflectivity (reflectance) R
refraction angle
resonant cavity
resonant frequency
Snell's laws
standing-wave ratio S
surface wave
total internal reflection
transmission coefficient τ
transmissivity (transmittance) T
transverse electric (TE) polarization
transverse magnetic (TM) polarization
unbounded-medium wavenumber
unpolarized
wavefront

在等式中设置 a = b = d 后：(8.124)，立方腔的 Q 表达式变为

$$Q = \frac{a}{3\delta_s} = \frac{1.68 \times 10^{-2}}{3 \times 5.89 \times 10^{-7}} \approx 9{,}500.$$

因此，腔体带宽为

$$\Delta f \approx \frac{f_{101}}{Q} \approx \frac{12.6 \times 10^9}{9{,}500} \approx 1.3 \text{ MHz}.$$

第8章总结

概念

- 描述平面电磁波在两种不同介质边界处的反射和传输行为的关系是满足 E 和 H 跨越边界的切向分量连续性条件的结果。
- 斯涅尔定律指出 $\theta_i = \theta_r$ 和

$$\sin\theta_t = (n_1/n_2)\sin\theta_i.$$

  对于$n_2 < n_1$这样的介质，当$\theta_i \geq \theta_c$时，入射波被边界全反射，其中θ_c是由$\theta_c = \sin^{-1}(n_2/n_1)$给出的临界角。
- 通过连续的多次反射，光可以通过光纤被引导。可以沿光纤传输的数字脉冲的最大数据速率由模态色散决定。
- 在给定偏振的布鲁斯特角下，入射波完全穿过边界传播。对于非磁性材料，布儒斯特角仅存在于平行极化。
- 任何入射到平面边界上的平面波都可以合成为垂直偏振波和平行偏振波之和。
- 传输线等效模型可用于表征不同介质之间边界的波传播、反射和传输。
- 波可以以横电（TE）和横磁（TM）模式的形式在金属波导中传播。对于每种模式，波导都有一个截止频率，低于该频率波就无法传播。
- 空腔谐振器可以支持特定谐振频率的驻波。

重要条款

提供以下术语的定义或解释其含义：

ω-β 图接收角 θ_a 入射角、反射角和透射角 布儒斯特角 θ_B 包层临界角 θ_c 截止频率 f_{mn} 截止波数 k_c 主模倏逝波 光纤芯掠入射群速度 u_g 折射率 n 模色散模式 光纤 平行偏振 垂直偏振 相位匹配条件 入射平面 偏振角 品质因数 Q 反射系数 Γ 反射率（反射率） R 折射角 谐振腔 谐振频率 斯涅尔定律 驻波比 S 表面波 全内反射 透射系数 τ 透射率（透射率） T 横向电 (TE) 偏振 横向磁 (TM) 偏振 无界介质波数 非偏振波前

Mathematical and Physical Models

Normal Incidence

$$\Gamma = \frac{E_0^{\mathrm{r}}}{E_0^{\mathrm{i}}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\tau = \frac{E_0^{\mathrm{t}}}{E_0^{\mathrm{i}}} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$\tau = 1 + \Gamma$$

$$\Gamma = \frac{\sqrt{\varepsilon_{\mathrm{r}_1}} - \sqrt{\varepsilon_{\mathrm{r}_2}}}{\sqrt{\varepsilon_{\mathrm{r}_1}} + \sqrt{\varepsilon_{\mathrm{r}_2}}} \quad (\text{if } \mu_1 = \mu_2)$$

Snell's Laws

$$\theta_{\mathrm{i}} = \theta_{\mathrm{r}}$$

$$\frac{\sin\theta_{\mathrm{t}}}{\sin\theta_{\mathrm{i}}} = \frac{u_{\mathrm{p}_2}}{u_{\mathrm{p}_1}} = \sqrt{\frac{\mu_1\varepsilon_1}{\mu_2\varepsilon_2}}$$

Oblique Incidence

Perpendicular Polarization

$$\Gamma_\perp = \frac{E_{\perp 0}^{\mathrm{r}}}{E_{\perp 0}^{\mathrm{i}}} = \frac{\eta_2\cos\theta_{\mathrm{i}} - \eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}} + \eta_1\cos\theta_{\mathrm{t}}}$$

$$\tau_\perp = \frac{E_{\perp 0}^{\mathrm{t}}}{E_{\perp 0}^{\mathrm{i}}} = \frac{2\eta_2\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{i}} + \eta_1\cos\theta_{\mathrm{t}}}$$

Parallel Polarization

$$\Gamma_\parallel = \frac{E_{\parallel 0}^{\mathrm{r}}}{E_{\parallel 0}^{\mathrm{i}}} = \frac{\eta_2\cos\theta_{\mathrm{t}} - \eta_1\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{t}} + \eta_1\cos\theta_{\mathrm{i}}}$$

$$\tau_\parallel = \frac{E_{\parallel 0}^{\mathrm{t}}}{E_{\parallel 0}^{\mathrm{i}}} = \frac{2\eta_2\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{t}} + \eta_1\cos\theta_{\mathrm{i}}}$$

Brewster Angle

$$\theta_{\mathrm{B}\parallel} = \sin^{-1}\sqrt{\frac{1}{1 + (\varepsilon_1/\varepsilon_2)}} = \tan^{-1}\sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

Waveguides

$$\beta = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

$$f_{mn} = \frac{u_{\mathrm{p}_0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$u_{\mathrm{p}} = \frac{\omega}{\beta} = \frac{u_{\mathrm{p}_0}}{\sqrt{1 - (f_{mn}/f)^2}}$$

$$u_{\mathrm{p}}u_{\mathrm{g}} = u_{\mathrm{p}_0}^2$$

$$Z_{\mathrm{TE}} = \frac{\eta}{\sqrt{1 - (f_{mn}/f)^2}}$$

$$Z_{\mathrm{TM}} = \eta\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}$$

Resonant Cavity

$$f_{mnp} = \frac{u_{\mathrm{p}_0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

$$Q \approx \frac{f_{mnp}}{\Delta f}$$

PROBLEMS

Section 8-1: Reflection and Transmission at Normal Incidence

*8.1 A plane wave in air with an electric field amplitude of 20 V/m is incident normally upon the surface of a lossless, nonmagnetic medium with $\varepsilon_{\mathrm{r}} = 25$. Determine:

(a) the reflection and transmission coefficients,

(b) the standing-wave ratio in the air medium, and

(c) the average power densities of the incident, reflected, and transmitted waves.

8.2 A plane wave traveling in medium 1 with $\varepsilon_{\mathrm{r}1} = 2.25$ is normally incident upon medium 2 with $\varepsilon_{\mathrm{r}2} = 4$. Both media are made of nonmagnetic, nonconducting materials. The electric field of the incident wave is given by

$$\mathbf{E}^{\mathrm{i}} = \hat{\mathbf{y}}8\cos(6\pi \times 10^9 t - 30\pi x) \quad (\mathrm{V/m}).$$

(a) Obtain time-domain expressions for the electric and magnetic fields in each of the two media.

* Answer(s) available in Appendix E.

数学与物理模型

垂直入射

$$\Gamma = \frac{E_0^{\mathrm{r}}}{E_0^{\mathrm{i}}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\tau = \frac{E_0^{\mathrm{t}}}{E_0^{\mathrm{i}}} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$\tau = 1 + \Gamma$$

$$\Gamma = \frac{\sqrt{\varepsilon_{\mathrm{r}_1}} - \sqrt{\varepsilon_{\mathrm{r}_2}}}{\sqrt{\varepsilon_{\mathrm{r}_1}} + \sqrt{\varepsilon_{\mathrm{r}_2}}} \quad (\text{if } \mu_1 = \mu_2)$$

斯涅尔定律

$$\theta_{\mathrm{i}} = \theta_{\mathrm{r}}$$

$$\frac{\sin\theta_{\mathrm{t}}}{\sin\theta_{\mathrm{i}}} = \frac{u_{\mathrm{p}_2}}{u_{\mathrm{p}_1}} = \sqrt{\frac{\mu_1\varepsilon_1}{\mu_2\varepsilon_2}}$$

斜入射

垂直极化

$$\Gamma_\perp = \frac{E_{\perp 0}^{\mathrm{r}}}{E_{\perp 0}^{\mathrm{i}}} = \frac{\eta_2\cos\theta_{\mathrm{i}} - \eta_1\cos\theta_{\mathrm{t}}}{\eta_2\cos\theta_{\mathrm{i}} + \eta_1\cos\theta_{\mathrm{t}}}$$

$$\tau_\perp = \frac{E_{\perp 0}^{\mathrm{t}}}{E_{\perp 0}^{\mathrm{i}}} = \frac{2\eta_2\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{i}} + \eta_1\cos\theta_{\mathrm{t}}}$$

平行极化

$$\Gamma_\parallel = \frac{E_{\parallel 0}^{\mathrm{r}}}{E_{\parallel 0}^{\mathrm{i}}} = \frac{\eta_2\cos\theta_{\mathrm{t}} - \eta_1\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{t}} + \eta_1\cos\theta_{\mathrm{i}}}$$

$$\tau_\parallel = \frac{E_{\parallel 0}^{\mathrm{t}}}{E_{\parallel 0}^{\mathrm{i}}} = \frac{2\eta_2\cos\theta_{\mathrm{i}}}{\eta_2\cos\theta_{\mathrm{t}} + \eta_1\cos\theta_{\mathrm{i}}}$$

布儒斯特角

$$\theta_{\mathrm{B}\parallel} = \sin^{-1}\sqrt{\frac{1}{1 + (\varepsilon_1/\varepsilon_2)}} = \tan^{-1}\sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

波导

$$\beta = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

$$f_{mn} = \frac{u_{\mathrm{p}_0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$u_{\mathrm{p}} = \frac{\omega}{\beta} = \frac{u_{\mathrm{p}_0}}{\sqrt{1 - (f_{mn}/f)^2}}$$

$$u_{\mathrm{p}}u_{\mathrm{g}} = u_{\mathrm{p}_0}^2$$

$$Z_{\mathrm{TE}} = \frac{\eta}{\sqrt{1 - (f_{mn}/f)^2}}$$

$$Z_{\mathrm{TM}} = \eta\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}$$

谐振腔

$$f_{mnp} = \frac{u_{\mathrm{p}_0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

$$Q \approx \frac{f_{mnp}}{\Delta f}$$

问题

第8-1节 垂直入射时的反射与透射

*8.1 空气中电场幅度为20 V/m的平面波，垂直入射到相对介电率 ε_{r}=25 的无损非磁性介质表面。求：

(a) 反射系数和透射系数，(b) 空气介质中的驻波比，以及

(c) 入射波、反射波和透射波的平均功率密度。

8.2 在 $\varepsilon_{\mathrm{r}1} = 2.25$ 的介质 1 中传播的平面波垂直入射到 $\varepsilon_{\mathrm{r}2} = 4$ 的介质 2 上。两种介质均由非磁性、非导电材料制成。入射波的电场由下式给出

$$\mathbf{E}^{\mathrm{i}} = \hat{\mathbf{y}}8\cos(6\pi \times 10^9 t - 30\pi x) \quad (\mathrm{V/m}).$$

(a) 获得两种介质中电场和磁场的时域表达式。

* 标记的习题在附录 E 提供答案。

(b) Determine the average power densities of the incident, reflected and transmitted waves.

8.3 A plane wave traveling in a medium with $\varepsilon_{r_1} = 9$ is normally incident upon a second medium with $\varepsilon_{r_2} = 4$. Both media are made of nonmagnetic, nonconducting materials. The magnetic field of the incident plane wave is given by

$$\mathbf{H}^i = \hat{\mathbf{z}}\, 2\cos(2\pi \times 10^9 t - ky) \qquad \text{(A/m)}.$$

(a) Obtain time-domain expressions for the electric and magnetic fields in each of the two media.

*(b) Determine the average power densities of the incident, reflected, and transmitted waves.

8.4 A 200 MHz, left-hand circularly polarized plane wave with an electric field modulus of 5 V/m is normally incident in air upon a dielectric medium with $\varepsilon_r = 4$ and occupies the region defined by $z \geq 0$.

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$.

(b) Calculate the reflection and transmission coefficients.

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted wave, and the total field in the region $z \leq 0$.

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium.

8.5 Repeat Problem 8.4, but replace the dielectric medium with a poor conductor characterized by $\varepsilon_r = 2.25$, $\mu_r = 1$, and $\sigma = 10^{-4}$ S/m.

8.6 A 50 MHz plane wave with electric field amplitude of 50 V/m is normally incident in air onto a semi-infinite, perfect dielectric medium with $\varepsilon_r = 36$. Determine:

*(a) Γ,

(b) the average power densities of the incident and reflected waves, and

(c) the distance in the air medium from the boundary to the nearest minimum of the electric field intensity, $|\mathbf{E}|$.

*8.7 What is the maximum amplitude of the total electric field in the air medium of Problem 8.6, and at what nearest distance from the boundary does it occur?

8.8 Repeat Problem 8.6, but replace the dielectric medium with a conductor with $\varepsilon_r = 1$, $\mu_r = 1$, and $\sigma = 2.78 \times 10^{-3}$ S/m.

*8.9 The three regions shown in **Fig. P8.9** contain perfect dielectrics. For a wave in medium 1, incident normally upon the boundary at $z = -d$, what combination of ε_{r_2} and d produces no reflection? Express your answers in terms of ε_{r_1}, ε_{r_3} and the oscillation frequency of the wave, f.

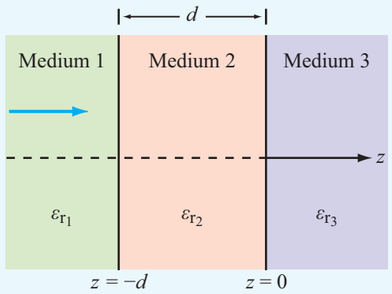


Figure P8.9 Dielectric layers for Problems 8.9 to 8.11.

8.10 For the configuration shown in **Fig. P8.9**, use transmission-line equations (or the Smith chart) to calculate the input impedance at $z = -d$ for $\varepsilon_{r_1} = 1$, $\varepsilon_{r_2} = 9$, $\varepsilon_{r_3} = 4$, $d = 1.2$ m, and $f = 50$ MHz. Also determine the fraction of the incident average power density reflected by the structure. Assume all media are lossless and nonmagnetic.

*8.11 Repeat Problem 8.10, but interchange ε_{r_1} and ε_{r_3}.

8.12 Orange light of wavelength 0.61 μm in air enters a block of glass with $\varepsilon_r = 1.44$. What color would it appear to a sensor embedded in the glass? The wavelength ranges of colors are violet (0.39 to 0.45 μm), blue (0.45 to 0.49 μm), green (0.49 to 0.58 μm), yellow (0.58 to 0.60 μm), orange (0.60 to 0.62 μm), and red (0.62 to 0.78 μm).

*8.13 A plane wave of unknown frequency is normally incident in air upon the surface of a perfect conductor. Using an electric-field meter, it was determined that the total electric field in the air medium is always zero when measured at a distance of 2 m from the conductor surface. Moreover, no such nulls were observed at distances closer to the conductor. What is the frequency of the incident wave?

8.14 Consider a thin film of soap in air under illumination by yellow light with $\lambda = 0.6$ μm in vacuum. The film is treated as a planar dielectric slab with $\varepsilon_r = 1.72$ and is surrounded on both sides by air. What film thickness would produce strong reflection of the yellow light at normal incidence?

*8.15 A 5 MHz plane wave with electric field amplitude of 10 (V/m) is normally incident in air onto the plane surface of a semi-infinite conducting material with $\varepsilon_r = 4$, $\mu_r = 1$,

(b) 确定入射波、反射波和透射波的平均功率密度。

8.3 在$\varepsilon_{r_1} = 9$ 的介质中传播的平面波垂直入射到$\varepsilon_{r_2} = 4$ 的第二介质上。两种介质均由非磁性、非导电材料制成。入射平面波的磁场由下式给出

$$\mathbf{H}^i = \hat{\mathbf{z}}\, 2\cos(2\pi \times 10^9 t - ky) \qquad \text{(A/m)}。$$

(a) 获得两种介质中电场和磁场的时域表达式。*(b) 确定入射波、反射波和透射波的平均功率密度。

8.4 电场模量为 5 V/m 的 200 MHz 左旋圆极化平面波垂直在空气中入射到 $\varepsilon_r = 4$ 的介电介质上，并占据由 $z \geq 0$ 定义的区域。

(a) 假设电场在 z = 0 和 t = 0 处为正最大值，写出入射波的电场相量表达式。(b) 计算反射和透射系数。(c) 写出反射波、透射波和 $z \leq 0$ 区域中总场的电场相量表达式。(d) 确定边界反射并传输到第二介质中的入射平均功率的百分比。

8.5 重复问题 8.4，但用特征为 $\varepsilon_r = 2.25$、$\mu_r = 1$ 和 $\sigma = 10^{-4}$ S/m 的不良导体替换介电介质。

8.6 电场幅度为 50 V/m 的 50 MHz 平面波垂直在空气中入射到 $\varepsilon_r = 36$ 的半无限理想介电介质上。确定：

*(a) Γ，(b) 入射波和反射波的平均功率密度，以及 (c) 空气介质中从边界到电场强度最接近最小值的距离，|E|。

*8.7 问题8.6 的空气介质中总电场的最大幅度是多少，它出现在距离边界最近的距离处？

8.8 重复问题 8.6，但将介电介质替换为具有 $\varepsilon_r = 1$、$\mu_r = 1$ 和 $\sigma = 2.78 \times 10^{-3}$ S/m 的导体。

*8.9 图 P8.9 所示的三个区域包含理想的电介质。对于介质 1 中的波，法线入射

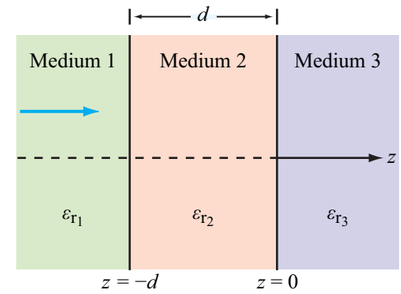


图 P8.9 问题 8.9 至 8.11 的介电层。

$z = -d$ 处的边界，ε_{r_2} 和 d 的哪种组合不会产生反射？用 ε_{r_1}、ε_{r_3} 和波的振荡频率 f 表达您的答案。

8.10 对于图 P8.9 所示的配置，使用传输线方程（或史密斯圆图）计算 $\varepsilon_{r_1} = 1$、$\varepsilon_{r_2} = 9$、$\varepsilon_{r_3} = 4$、d = 1.2 m 和 f = 50 MHz 时 $z = -d$ 处的输入阻抗。还确定结构反射的入射平均功率密度的分数。假设所有介质都是无损且非磁性的。

*8.11 重复问题 8.10，但互换 ε_{r_1} 和 ε_{r_3}。

8.12 波长0.61的橙光μ 空气中的 m 进入一块玻璃ε_r =1.44 。嵌入玻璃中的传感器会呈现什么颜色？颜色的波长范围是紫色（0.39 至 0.45μ 米），蓝色（0.45 至 0.49μ 米），绿色（0.49 至 0.58μ 米），黄色（0.58 至 0.60μ 米），橙色（0.60 至 0.62μ m）和红色（0.62 至 0.78μ 米）。

*8.13 频率未知的平面波垂直在空气中入射到理想导体的表面。使用电场计，确定在距离导体表面 2 m 处测量时，空气介质中的总电场始终为零。此外，在距离导体更近的地方没有观察到这样的零点。入射波的频率是多少？

8.14 考虑真空中 $\lambda = 0.6$ μ m 黄光照射下空气中的一层肥皂薄膜。该薄膜被视为 $\varepsilon_r = 1.72$ 的平面介电板，并且两侧都被空气包围。多大的薄膜厚度会在法向入射时产生强烈的黄光反射？

*8.15 电场幅度为 10 (V/m) 的 5 MHz 平面波垂直在空气中入射到 $\varepsilon_r = 4$、$\mu_r = 1$ 的半无限导电材料的平面上，

and $\sigma = 100$ (S/m). Determine the average power dissipated (lost) per unit cross-sectional area in a 2 mm penetration of the conducting medium.

8.16 A 0.5 MHz antenna carried by an airplane flying over the ocean surface generates a wave that approaches the water surface in the form of a normally incident plane wave with an electric-field amplitude of 3,000 (V/m). Seawater is characterized by $\varepsilon_r = 72$, $\mu_r = 1$, and $\sigma = 4$ (S/m). The plane is trying to communicate a message to a submarine submerged at a depth d below the water surface. If the submarine's receiver requires a minimum signal amplitude of 0.01 (μV/m), what is the maximum depth d where successful communication is still possible?

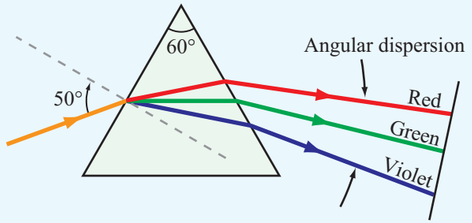


Figure P8.18 Prism of Problem 8.18.

Sections 8-2 and 8-3: Snell's Laws and Fiber Optics

*8.17 A light ray is incident on a prism in air at an angle θ, as shown in Fig. P8.17. The ray is refracted at the first surface and again at the second surface. In terms of the apex angle ϕ of the prism and its index of refraction n, determine the smallest value of θ for which the ray will emerge from the other side. Find this minimum θ for $n = 1.4$ and $\phi = 60°$.

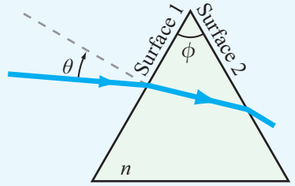


Figure P8.17 Prism of Problem 8.17.

8.18 For some types of glass, the index of refraction varies with wavelength. A prism is made of a material with

$$n = 1.71 - \frac{4}{30}\lambda_0 \qquad (\lambda_0 \text{ in } \mu\text{m}),$$

where λ_0 is the wavelength in vacuum, and is used to disperse white light, as shown in Fig. P8.18. The white light is incident at an angle of 50°, the wavelength λ_0 of red light is 0.7 μm, and that of violet light is 0.4 μm. Determine the angular dispersion in degrees.

*8.19 The two prisms in Fig. P8.19 are made of glass with $n = 1.5$. What fraction of the power density carried by the ray incident upon the top prism emerges from the bottom prism? Neglect multiple internal reflections.

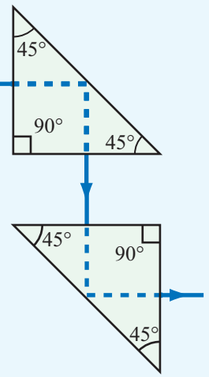


Figure P8.19 Periscope prisms of Problem 8.19.

8.20 A parallel-polarized plane wave is incident from air at an angle $\theta_i = 30°$ onto a pair of dielectric layers, as shown in Fig. P8.20.

(a) Determine the angles of transmission θ_2, θ_3, and θ_4.

(b) Determine the lateral distance d.

8.21 A light ray incident at 45° passes through two dielectric materials with the indices of refraction and thicknesses given in Fig. P8.21. If the ray strikes the surface of the first dielectric at a height of 2 cm, at what height will it strike the screen?

*8.22 Figure P8.22 depicts a beaker containing a block of glass on the bottom and water over it. The glass block contains a small air bubble at an unknown depth below the water surface. When viewed from above at an angle of 60°, the air bubble appears at a depth of 6.81 cm. What is the true depth of the air bubble?

和σ =100（S/m）。确定导电介质渗透 2 毫米时每单位横截面积消耗（损失）的平均功率。

8.16 飞越海洋表面的飞机携带的 0.5 MHz 天线产生的波以垂直入射平面波的形式接近水面，电场幅度为 3,000（V/m）。海水的特征为$\varepsilon_r = 72$、$\mu_r = 1$ 和$\sigma = 4$ (S/m)。这架飞机正试图向潜入水面以下深度 d 的潜艇传达信息。如果潜艇的接收器要求的最小信号幅度为 0.01 (μ V/m)，那么仍然可以成功通信的最大深度 d 是多少?

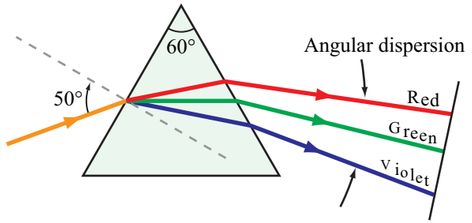


图 P8.18 问题 8.18 的棱镜。

第 8-2 节和 8-3 节：斯涅尔定律和光纤

*8.17 光线以θ角度入射到空气中的棱镜上，如图P8.17所示。光线在第一表面折射，并在第二表面再次折射。根据棱镜的顶角ϕ及其折射率n，确定光线从另一侧射出的θ的最小值。求 n = 1.4 和 ϕ = 60° 的最小值 θ。

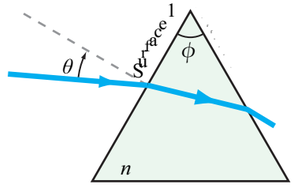


图 P8.17 问题 8.17 的棱镜。

8.18 对于某些类型的玻璃，折射率随波长而变化。棱镜由以下材料制成

$$n = 1.71 - \frac{4}{30}\lambda_0 \qquad (\lambda_0 \text{ in } \mu\text{m}),$$

其中λ_0为真空中的波长，用于色散白光，如图P8.18所示。白光以50°的角度入射，红光的波长λ_0为0.7μ m，紫光的波长为0.4μ m。确定角色散（以度为单位）。

*8.19 图 P8.19 中的两个棱镜由 n = 1.5 的玻璃制成。入射到顶部棱镜的光线所携带的功率密度从底部棱镜射出的功率密度的几分之几？忽略多重内部反射。

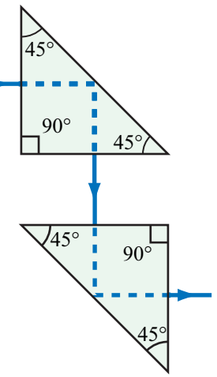


图 P8.19 问题 8.19 的潜望镜棱镜。

8.20 平行偏振平面波从空气以$\theta_i = 30°$角度入射到一对介电层上，如图P8.20所示。

(a) 确定传输角 θ_2、θ_3 和 θ_4。 (b) 确定横向距离 d。

8.21 以 45° 入射的光线穿过两种介电材料，其折射率和厚度如图 P8.21 所示。如果射线以 2 厘米的高度撞击第一个电介质的表面，那么它会以什么高度撞击屏幕?

*8.22 图 P8.22 描绘了一个烧杯，底部装有一块玻璃，上面有水。玻璃块在水面以下未知深度处包含一个小气泡。当从上方以 60° 角度观察时，气泡出现的深度为 6.81 厘米。气泡的真实深度是多少?